

Compare GPT-3.5 Turbo and GPT-4 Turbo at Lowest word-level confidence experiment

Maryam Naderi

Idea

Based on last experiment that we consider sentence-level confidence, Mathew suggested me to do word-level confidence experiment.

Enno suggested me that it could be easier to start with lowest word-level confidence.

For the moment, I did two experiments with 2 model: GPT-4 Turbo and GPT-3.5 Turbo without considering lowest word-level confidence.

Results for GPT-4 Turbo vs GPT-3.5 Turbo

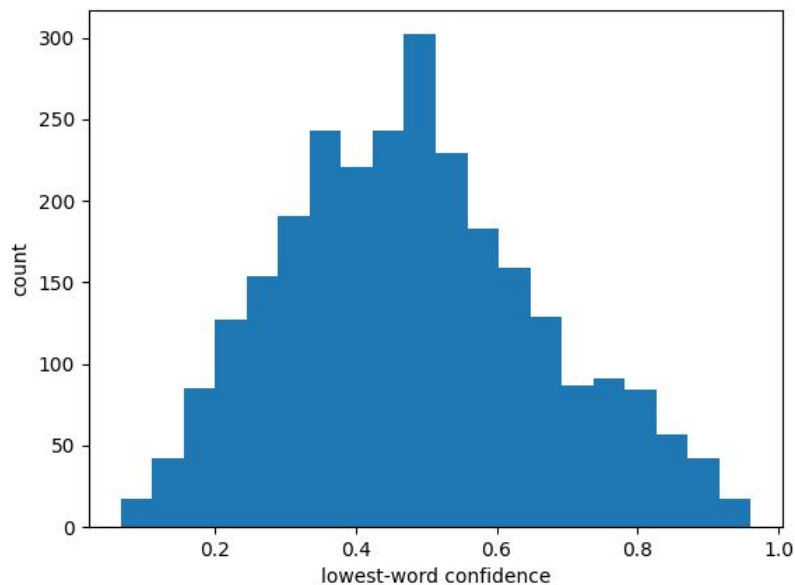
GPT-4 Turbo

WER_original:	8.6116%
WER_corrected_chatgpt:	6.9935%
CER_original:	3.5707%
CER_corrected_chatgpt:	3.4570%
SER_original:	61.4502%
SER_corrected_chatgpt:	56.0488%

GPT-3.5 Turbo

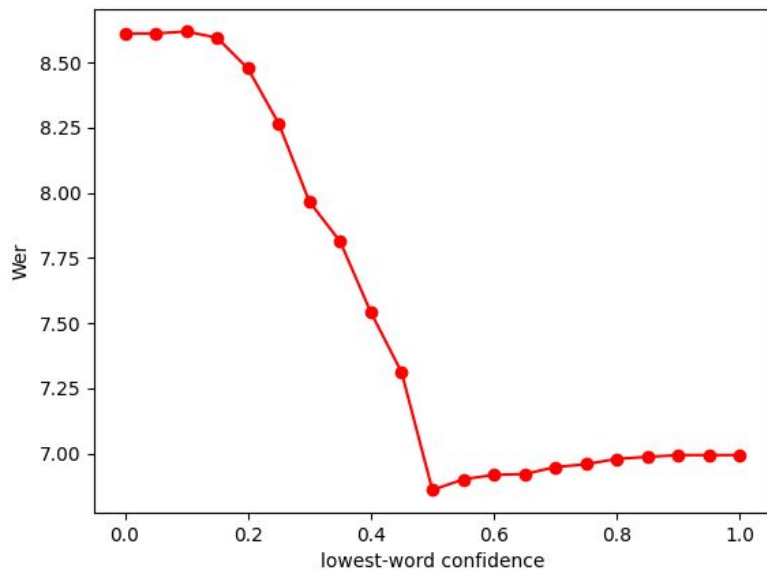
WER_original:	8.6116%
WER_corrected_chatgpt:	6.8042%
CER_original:	3.5707%
CER_corrected_chatgpt:	3.1879%
SER_original:	61.4502%
SER_corrected_chatgpt:	54.7170%

Histograms for GPT-4 Turbo and GPT-3.5 Turbo

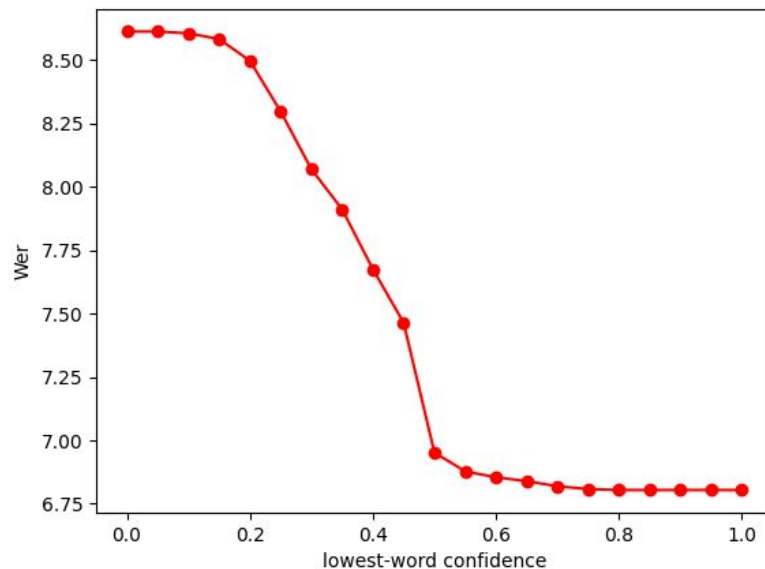


Wer plots for GPT-4 Turbo and GPT-3.5 Turbo

GPT-4 Turbo



GPT-3.5 Turbo



Next step

Find the best threshold for confidence and just call ChatGPT for ASR input that have the lowest word-level confidence less than threshold.

Thank you